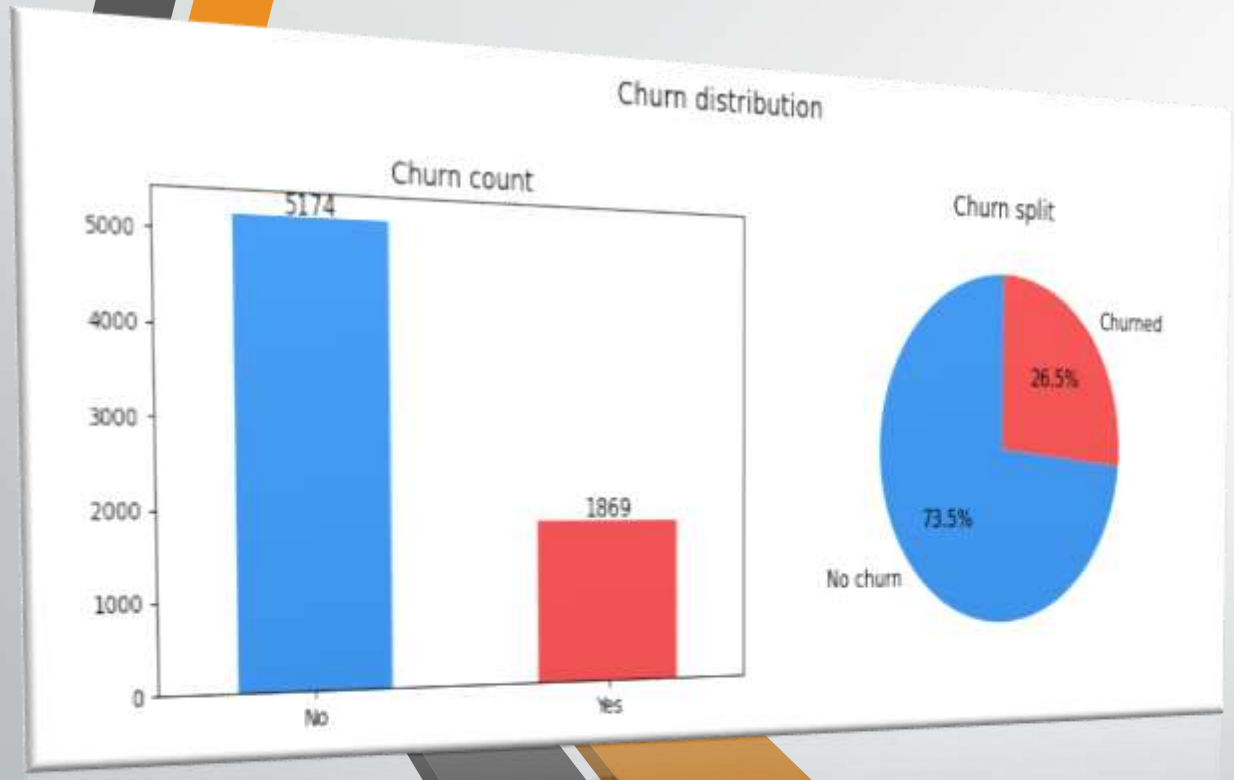
Customer Churn Analysis

A telecom company is losing customers but doesn't know why. This project identifies who is likely to churn, why they churn, and what actions the business should take — backed by data

Why Are We Losing Customers?



- A data-driven churn analysis for a telecom company

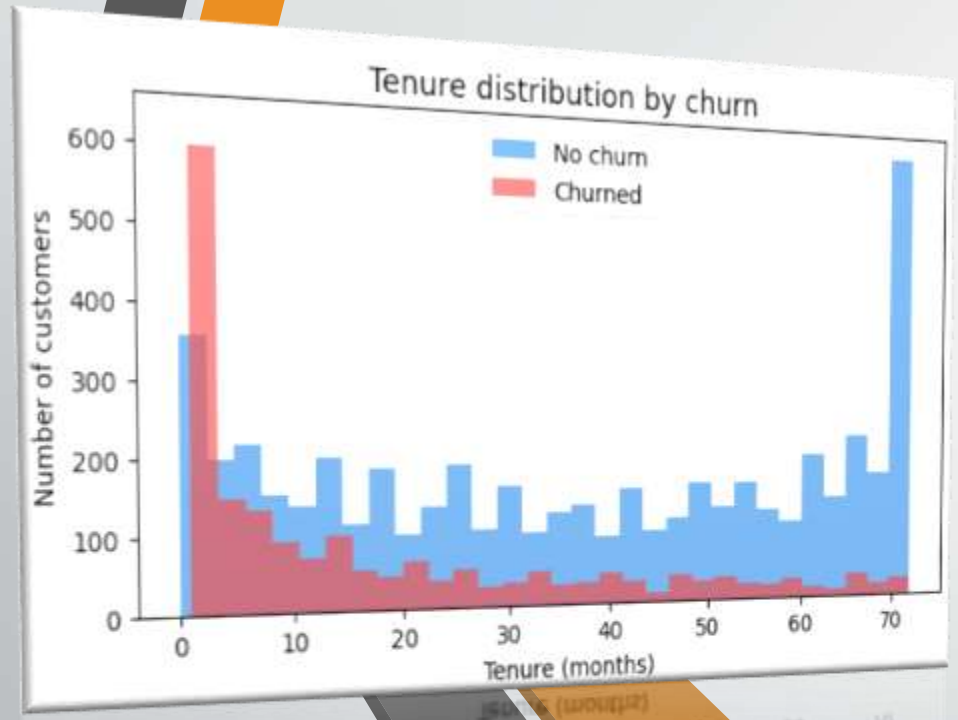
26.5% of customers churned last year

The business didn't know:

- Who was about to leave
- Why they were leaving
- What to do about it

This project answers all three questions.

What I Analyzed



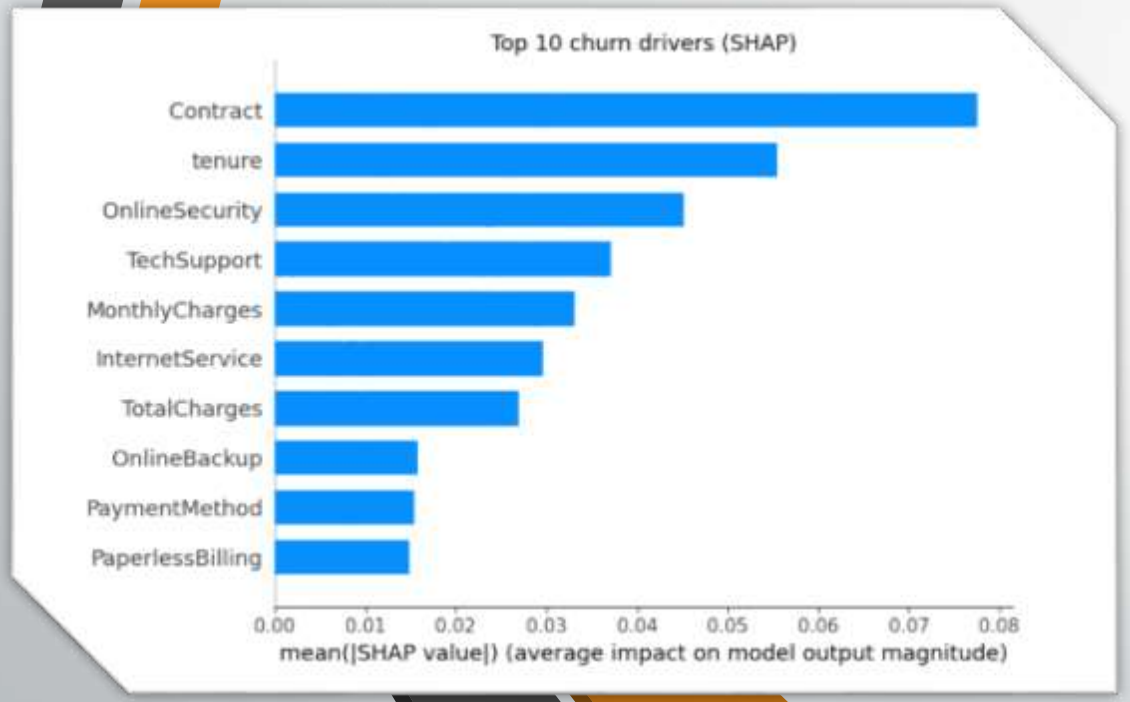
Dataset: 7,043 customers · 21 features · Kaggle (Telco Churn)

Tools: Python · scikit-learn · SHAP · Power BI

Approach:

1. Exploratory analysis — understand who churns
2. Prediction model — score every customer 0 to 1
3. Root cause analysis — explain why using SHAP
4. Churn playbook — turn findings into actions

3 Root Causes of Churn



1. Contract type

Month-to-month customers churn at 43%
vs 11% for annual contracts

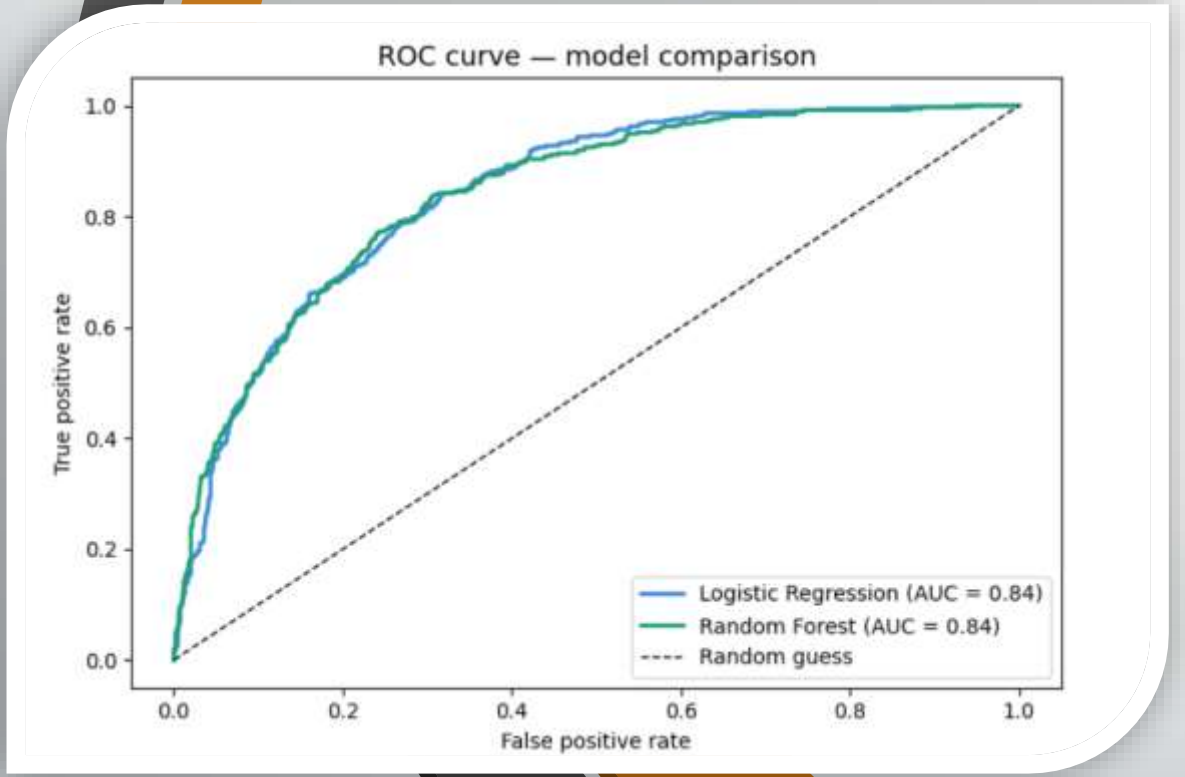
2. Early tenure is critical

Most churn happens in the first 12 months
Customers who stay past 2 years rarely leave

3. High charges + no support = highest risk

Customers paying $70+/month without tech support
have a churn probability of over 70%

Predicting Who Will Leave Next



Model: Random Forest Classifier

ROC-AUC: 84.0% ← catches 84% of churners

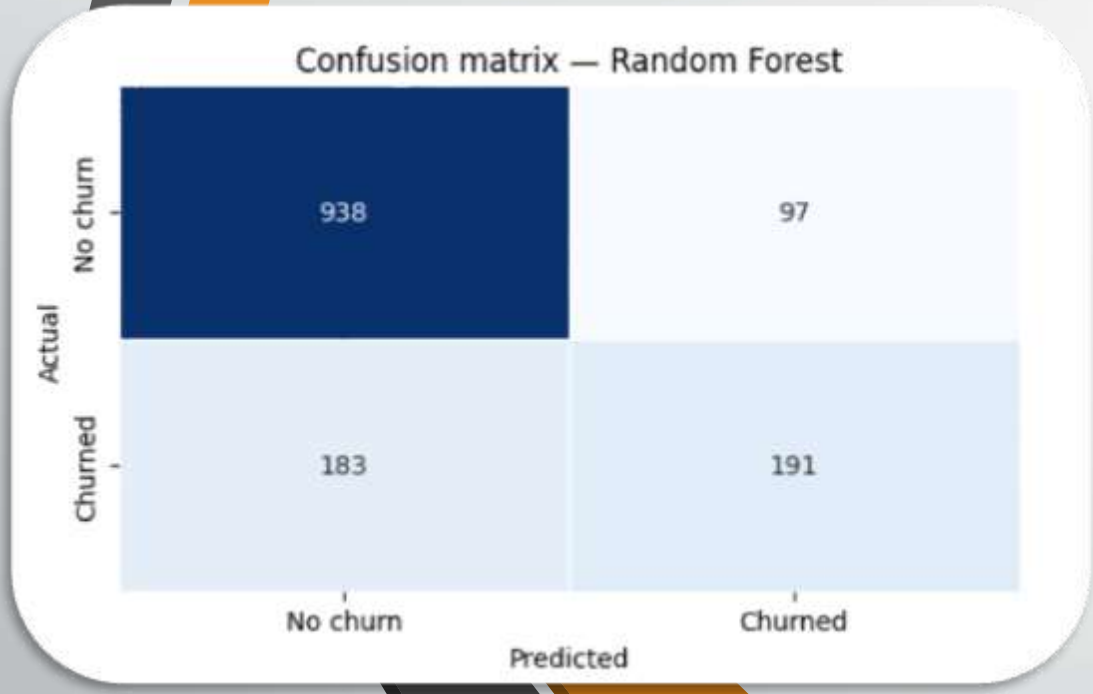
Output: Every customer gets a churn score from 0 to 1

High risk (score > 0.6) → 1,455 customers

Medium risk (score > 0.3) → 1,486 customers

Low risk (score < 0.3) → 4,102 customers

From Insight to Action

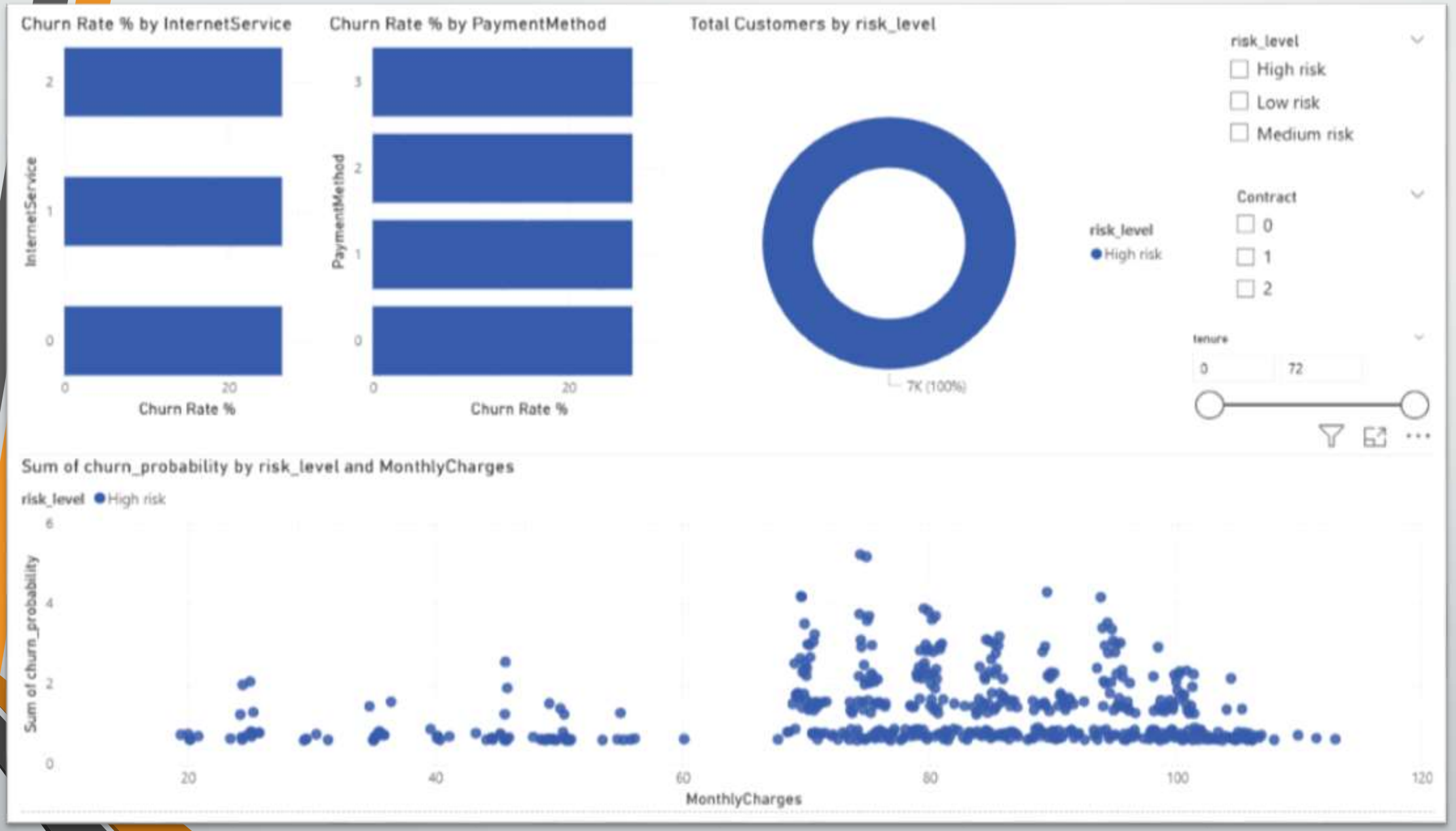


Churn Playbook — 5 trigger rules:

Score > 0.8 → Account manager call (24 hrs)
Score > 0.6 → Retention email (48 hrs)
Inactive 7 days → Re-engagement email
Month-to-month + risk → Offer annual plan discount
Tenure < 30 days → Onboarding check-in

Estimated impact:
1,455 high-risk customers identified
20% retention rate = ~$1,140 MRR recovered/month

Tools: Python · Power BI · SHAP · GitHub


Churn Rate % by InternetService
InternetService
2
1
0
0
20
Churn Rate %
Churn Rate % by PaymentMethod
PaymentMethod
3
2
1
0
0
20
Churn Rate %
Total Customers by risk_level
7K (100%)
risk_level
High risk
risk_level
High risk
Low risk
Medium risk
Contract
0
1
2
tenure
0
72
Sum of churn_probability by risk_level and MonthlyCharges
risk_level
High risk
Sum of churn_probability
6
4
2
0
20
40
60
80
100
120
MonthlyCharges

THANK YOU !